Lab 7 Fuzz Testing

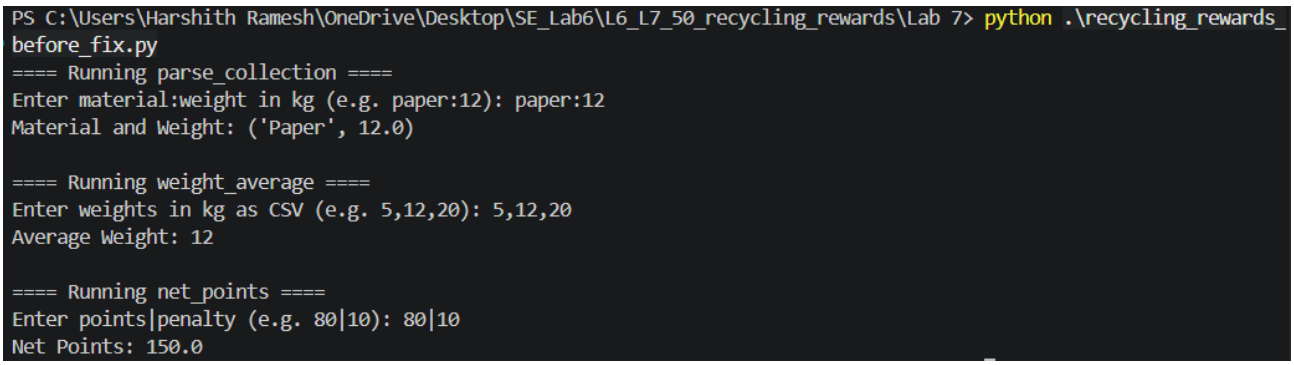
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Step 2 Original application run

This runs the preserved original application before the fixes. At its three prompts, enter the sample inputs shown below so the output is easy to compare.

```
python .\recycling_rewards_before_fix.py
Inputs when prompted: paper:12   then   5,12,20   then   80|10
```

Screenshot 1 Original application output



Step 4 Fuzz tests against the original application

```
Copy-Item .\recycling_rewards.py .\recycling_rewards_fixed_tmp.py
Copy-Item .\recycling_rewards_before_fix.py .\recycling_rewards.py -Force
python -m pytest -q --tb=no test_recycling_rewards.py
Copy-Item .\recycling_rewards_fixed_tmp.py .\recycling_rewards.py -Force
Remove-Item .\recycling_rewards_fixed_tmp.py
```

Screenshot 2 Failing fuzz tests on the original code

```
PS C:\Users\Harshith Ramesh\OneDrive\Desktop\SE_Lab6\L6_L7_50_recycling_rewards\Lab 7> Copy-Item .\recycling_rewar
ds.py .\recycling_rewards_fixed_tmp.py
PS C:\Users\Harshith Ramesh\OneDrive\Desktop\SE_Lab6\L6_L7_50_recycling_rewards\Lab 7> Copy-Item .\recycling_rewar
ds_before_fix.py .\recycling_rewards.py -Force
PS C:\Users\Harshith Ramesh\OneDrive\Desktop\SE_Lab6\L6_L7_50_recycling_rewards\Lab 7> python -m pytest -q --tb=no
 test_recycling_rewards.py
FFFFF.                                                                   [100%]
========================================= short test summary info =========================================
FAILED test_recycling_rewards.py::test_parse_collection_no_crash - BaseExceptionGroup: Hypothesis found 3 distinct
 failures. (3 sub-exceptions)
FAILED test_recycling_rewards.py::test_weight_average_no_crash - BaseExceptionGroup: Hypothesis found 2 distinct f
ailures. (2 sub-exceptions)
FAILED test_recycling_rewards.py::test_net_points_no_crash - BaseExceptionGroup: Hypothesis found 3 distinct failu
res. (3 sub-exceptions)
FAILED test_recycling_rewards.py::test_parse_collection_keeps_valid_input_behavior - ValueError: could not convert
 string to float: 'not-a-number'
FAILED test_recycling_rewards.py::test_weight_average_keeps_valid_input_behavior - ValueError: invalid literal for
 int() with base 10: ''
5 failed, 1 passed in 1.57s
```

Step 5 Fixed application run

```
python .\recycling_rewards.py
Inputs when prompted: paper:12   then   5,12,20   then   80|10
```

Screenshot 3 Fixed application output

```
PS C:\Users\Harshith Ramesh\OneDrive\Desktop\SE_Lab6\L6_L7_50_recycling_rewards\Lab 7> python .\recycling_rewards.
py
==== Running parse_collection ====
Enter material:weight in kg (e.g. paper:12): paper:12
Material and Weight: ('Paper', 12.0)

==== Running weight_average ====
Enter weights in kg as CSV (e.g. 5,12,20): 5,12,20
Average Weight: 12

==== Running net_points ====
Enter points|penalty (e.g. 80|10): 80|10
Net Points: 150.0
```

Step 6 Fuzz tests on the fixed application

```
python -m pytest -q test_recycling_rewards.py
```

Screenshot 4 Passing fuzz tests

```
PS C:\Users\Harshith Ramesh\OneDrive\Desktop\SE_Lab6\L6_L7_50_recycling_rewards\Lab 7> python -m pytest -q test_re
cycling_rewards.py
......                                                                   [100%]
6 passed in 0.42s
```